Sai Rithwik M

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Education

Integrated M. Tech | ***Computer Science and Engineering*** August 2018 – July 2023
International Institute of Information Technology, Bangalore Bengaluru, India

Experience

Software Development Engineer - 2 | ***Cred*** December 2025 – Present
- Scaled the Autopay platform from **5K to 100K daily transactions**, supporting a TPV of **$21M/month**.
- Managed core infrastructure across diverse business lines (Credit Cards, Utilities, Gold SIPs) and successfully drove the **0-to-1 launch** of Wallet and Mobile Recharge auto-payments. Slashed the onboarding time for new business lines from **multiple sprints to just 2 days**.
- Led multiple reliability engineering initiatives, driving a **58% reduction in daily payment failures** (12K to 5K)
  - **Architecture & Resilience:** Migrated multi-day payment workloads from in-memory orchestrator to Temporal and introduced circuit breaker patterns to prevent cascading downstream outages. Reduced incidents by **75%**
  - **Observability:** Engineered end-to-end payment telemetry by defining **L0, L1, L2 metrics**, capturing metrics using prometheus, setting up alerts, and building recon automations incase of failures.

Software Development Engineer - 1, 2 | ***Navi*** July 2023 – December, 2025
- Worked with Debt Management Platforms and Calling Teams to build systems for debt recovery operations.
- Built a prioritization platform that takes various signals to recommend the best cases for debt recovery agents.
- Enabled multi-tenancy of debt management platforms for both Lending and Credit charters.
- Developed internal tools and maintained CRM which helps agents improve their performance and productivity.
- Recognized with **"Game Changer"** (JFM '25) for enabling collections using AI voice bot and **"Spearhead"** (May '24) for building the tele-calling case allocation layer.

Projects

Debt Recovery Prioritization Platform | ***Elasticsearch, Kotlin, Spring Boot*** September 2024
- Built an intelligent prioritization platform that analyzes multiple signals including customer behavior, payment history, and risk scores to recommend optimal cases for debt recovery agents.
- The time and cost to onboard agents also reduced from 2 weeks to 1 week due to the optimised workload.
- Optimised Kafka ingestion pipelines to improve write throughput by parallely processing messages in queue utilising CPU along with Kotlin co-routines over increasing partitions, thereby saving cost.

Optimal Agent-Case Allocation System | ***Kotlin, Spring Boot, Databricks*** March 2024
- Designed and developed an end-to-end allocation system for optimal distribution of cases to recovery agents based on their languages, their EMI caseload constraints and many other factors.
- Orchestrated Linear Programming (LPP) models to optimize agent assignments considering caseload levels, agent specializations, geographical constraints, and performance metrics.
- Owned this business-critical feature that streamlines operations and recovers over **Rs. 100 crores per month**

Global Search Engine | ***Elasticsearch, Java, Spring Boot*** January 2024
- Built a global search platform enabling agents to search cases by names, case information with **100 ms** latency.
- Implemented fuzzy search, shard routing for data consistency, and migration strategies for schema changes.
- Improved agent productivity by reducing search operations from **3 clicks to 1 click**

Skills

Programming Languages: Java, Kotlin, Python
Tools: Elasticsearch, Postgres, DynamoDB, Kafka, SQS, Jenkins, K8s
Languages: English (C1), Hindi (C1), Telugu (Native), Tamil (B1)

Community Involvement

Co-Licensee, TEDxIIITBangalore 2020 – 2021
- Co-Licensee of TEDxIIITBangalore an annual event where speakers from various disciplines share ideas.
- Organised the first virtual edition of TEDxIIITBangalore which gained over **75k views** on Youtube.